

PRECISION CARE CHALLENGE 2026 · PHASE 1, IDEA SUBMISSION

ClaimCast

Know what your hospital bill will actually cost you. **Before** you are admitted, not **after** you are discharged.

TRACK: HOSPITALITY · Holistic Optimization System for Policy-Integrated Admission & Treatment Intelligence

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Explain The Problem You Are Attempting to Solve

India's most expensive healthcare decision is made in ninety seconds, at an admission counter, by someone who has never read their policy.

IRDAI's chairman has said claim payouts often fall short of what policyholders expect.

AJAY SETH, CHAIRMAN, IRDAI · NOV 2025

The regulator has named the gap. Policy bots explain the wording; none of them price the decision in front of you.

Proportionate deduction · IRDAI/HLT/REG/CIR/151/06/2020 · maximal case; scope varies by policy wording

THE ROOM-RENT TRAP

50% of the bill can be refused

Your policy caps the room at **₹5,000** a day. The hospital gave you one at **₹10,000**. Surgeon, OT, anesthesia, nursing: **every charge your policy ties to room category** is scaled down by that same ratio.

₹39,085

mean out-of-pocket per cancer hospitalization
12,148-patient multi-centre Indian study

₹3.31 L

out-of-pocket per cancer patient, per year
diagnostics 36% · medicines 45%

Cost shock drives distress financing, delayed care and outright **treatment abandonment**. It decides whether treatment finishes.

2

Problems solved

BASE BILL ESTIMATION

Forecasts your itemized bill from public tariff data, before you're admitted.

INSURANCE ANALYSIS

Computes exactly what your policy refuses, line by line, cited to the clause.

Briefly Explain Your Idea/Solution

Insurers compute what you are owed **after discharge**. ClaimCast computes it **before admission**.

Same arithmetic. Different timestamp. That is the whole idea.

1 · ADMISSION

Which hospital

gap between in-network and out-of-network payout

₹1.2 L

at stake

2 · ADMISSION

Which room class

private room vs. semi-private, after proportionate deduction

₹78,400

at stake

3 · INVESTIGATION

Cashless or reimburse

cash you must float upfront if not cashless

₹2 L

at stake

4 · PROCEDURE

Implant / consumable

cost of implants excluded under List I

₹45,000

at stake

5 · RECOVERY

24-hour admission rule

lost if care is billed day-care, not in-patient

₹22,000

at stake

THE CARE JOURNEY THE PROBLEM STATEMENT NAMES

we price the decision at every stage, and each fork closes permanently the moment it passes

CONSTRAINT OPTIMIZATION

The model re-runs your claim across every legal alternative and ranks them by what you keep, filtering out anything that breaks a constraint — Room-rent cap **₹10,000** , waiting period **1 yr**.

IS THIS UNIQUE, OR AN IMPROVISATION?

Unique. Existing tools **retrieve** policy text. ClaimCast **simulates the claim and optimizes the decision**. Retrieval → Simulation.

YOU WALK IN KNOWING WHAT YOU WILL OWE. Same admission, ₹78,400 less out of pocket, because the caregiver was told before the decision closed.

Explain Your Solution

We estimate what you'll be billed. We guarantee what won't be paid.

AI where uncertain.
Arithmetic where known.
The LLM never decides the money.

TWO HALVES

A · FORECAST

Predicts the itemised bill from published tariff data. Uncertain, so it returns a range.

B · ADJUDICATION

Computes what the policy refuses, line by line. Exact, with every ₹ cited to its clause.

COUNTERFACTUAL Re-runs the claim for every legal alternative and ranks them by what you keep.

AGAINST THE CONVENTIONAL ALTERNATIVES

	Hospital / TPA desk	Policy bots · Ditto/PB	ClaimCast
Speaks	after admission	any time, no context	at each decision
Returns	a verbal estimate	a text summary	itemized ₹ + clause
Acts for	the hospital	no one	you

PROS

- Deterministic and auditable
- Runs entirely on public data
- Rule-packs: private, PM-JAY, ESI

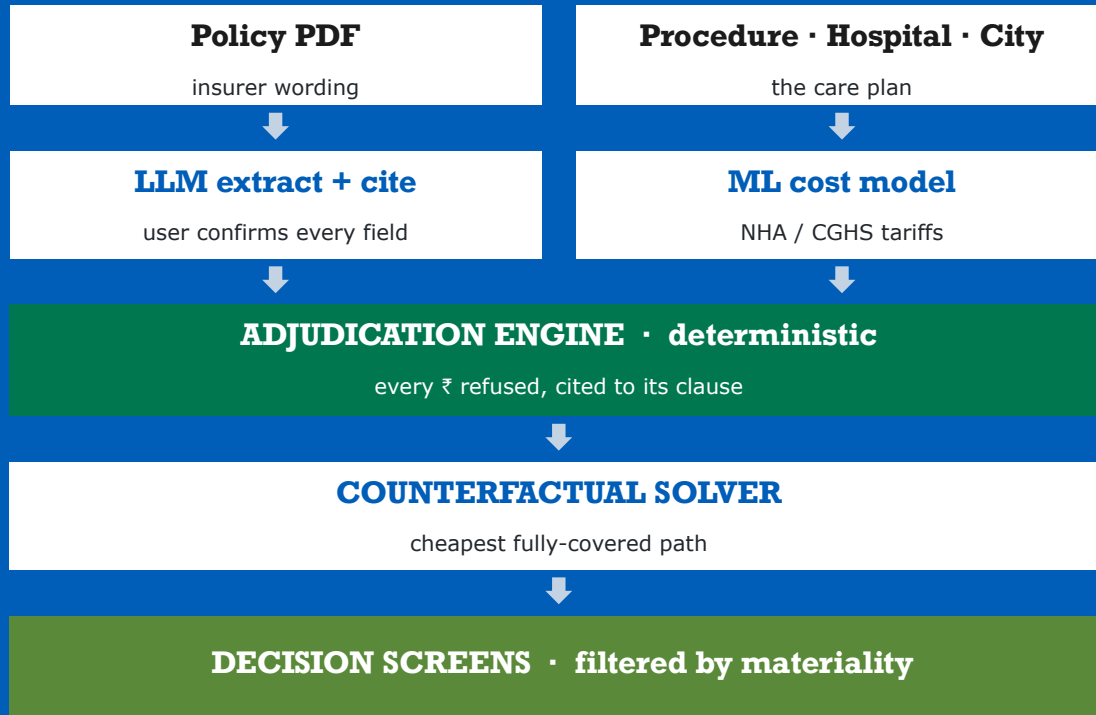
CONS

- Forecast bounded by published tariffs
- Needs the policy document up front
- An estimate, never a guarantee

ClaimCast	DECISION 2 OF 5 · ROOM CLASS · STAGE: ADMISSION	PROTOTYPE SCREEN
○ Private room • Semi-private	₹10,000 / day ₹5,000 / day your share ₹1,26,900 your share ₹48,500 ← recommended by ClaimCast	
✓ You keep ₹78,400	Room Rent Sub-limit, Clause 3.2 · proportionate deduction on the associated charges	

Architecture & AI Pipeline

ARCHITECTURE



ML cost model

XGBoost quantile regression — predicts a cost range, not a point estimate
Trained on: NHA/PM-JAY rates ($\approx 1,900$ procedures) · CGHS city-wise tariffs · past forecast-vs-actual outcomes (self-learning loop)

LLM extract + cite

GPT-4o / Claude, RAG over the uploaded policy PDF
Extracts: sum insured, sub-limits, co-pay %, exclusions, waiting periods — each field cited to its source clause
Extraction only, never adjudication — a misread field fails safe, flagged for user confirmation, never silently changes what's refused

MEMORY · SELF-LEARNING LOOP

Logs (procedure, hospital, city, forecast, actual billed amount) per case. Retrains the cost model on the combined historical + incoming pool — the range narrows as volume grows.

WHAT WE BUILD, AND WHAT WE SIMULATE

Build

Policy → constraint parser

Extracts sub-limits, co-pay, and exclusions from the uploaded policy PDF into structured rules. Every extracted field is user-confirmed before it's used.

Adjudication + deduction engine

Applies the parsed rules deterministically to compute what's refused, line by line. No LLM in the money path — output is auditable.

Cost model on NHA / CGHS rates

Forecasts the itemized bill from public tariff data for the given procedure, hospital, and city. Returns a range, not a point estimate.

Counterfactual solver + screens

Re-runs the claim across every legal alternative and ranks them by what the user keeps. Renders as the five decision screens.

Memory

Stores outcomes from every claim processed. Used to retrain the cost model over time, so estimates sharpen with usage.

Simulate

Live hospital tariff feed

Real hospital billing rates by procedure and room class. Needs a hospital partner to open; approximated with public data until then.

Insurer cashless API / HIS bill feed

Real-time cashless approval status and hospital billing system data. Needs an insurer/hospital integration; not available pre-launch.

DATA WE SOURCE OURSELVES · ALL PUBLIC

- NHA / PM-JAY Health Benefit Packages (≈1,900 procedure rates)
- CGHS city-wise rate lists · NHA empanelled-hospital registry
- IRDAI standard wordings (Arogya Sanjeevani) · IRDAI List I non-payables

The two simulated feeds need an insurer and a hospital to open them, not more engineering.

REFERENCES

1. IRDAI, Circular IRDAI/HLT/REG/CIR/151/06/2020, room-rent proportionate deduction.
2. Financial toxicity of cancer treatment in India, Front. Public Health, 2023 (n = 12,148).
3. Catastrophic expenditure & treatment attrition, Lancet Reg. Health SE Asia, 2022;6:100058.
4. IRDAI Master Circular on Health Insurance Business, 2024.

SAFE BY CONSTRUCTION every field is user-confirmed · outputs are labelled estimates, never guarantees · no medical advice, money only · no identifiable patient data